MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 8315

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pulsed amplifier, mechanically tunable, frequency range 5,385 to 5,900 MHz

ABSOLUTE RATINGS:

Parameter:	Ef	If(surge)	eb	ib	pi	Pi	pd	Pd	Po	Du
Unit:	V	A	kv	a	kw	W	w	W	W	---
Maximum:	6.9	8.0	9.0	0.95	8.55	17	10	10	10	0.002
Minimum:	5.7	---	---	---	---	---	---	---	---	---
	(Note 1)		(Note 2)							(Note 3)

ABSOLUTE RATINGS:

				Operational temperature			Tuner	
Parameter:	tp	tk	TA	Cathode seal	Body	Collector	stop torque	Alt
Unit:	µs	sec	°C	°C	°C	°C	in-lbs	ft
Maximum:	5.0	---	+85	+125	+100	+150	6	10,000
Minimum:	0.1	120	-21	---	---	---	---	---
	(Note 3)	(Note 1)		(Note 4)	(Note 4)	(Note 4)	(Note 5)	

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Start-up procedure:	See note 1
Mounting position:	Any	Cooling:	Forced air (see note 4)
Connectors:	M39012/16-0001 (See figure 1)	Weight:	3.25 lbs (approx)
		Tuning:	See note 5; and mechanical tuning characteristics (listed under QCI, part 1)
Ground:	Tube shell		
Base:	8-pin octal		

TEST CONDITIONS:

Parameter:	Ef	eb	pd	tp	tr	tf	σ	σ'	TA	F
Unit:	V	kv	mw	µs	µs	µs	---	---	°C	MHz
Tolerance:	---	---	Max	±0.3	±0.2	±0.3	Max	---	±28	---
Test condition 1:	6.3	7.0	0	3.0	0.6	0.5	2:1	1.5:1	60	---
Test condition 2:	6.3	7.0	50	3.0	0.6	0.5	2:1	1.5:1	60	F1, F2, F3
Test condition 3:	6.3	7.0	500	3.0	0.6	0.5	2:1	1.5:1	60	F1, F2, F3
	(Note 1)	(Note 2)		(Note 3)	(Note 3)	(Note 3)			(Note 13)	(Notes 5,11, 12)

GENERAL:

Qualification - Required
Shelf life - 12 months (see note 20)

Frequency	
F	MHz
1	5,385
2	5,650
3	5,900

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
1047	High-and low-temperature performance (operating)	16	3	TA = 30° to 85°C	Δpo	---	4.0	dB
1031	Vibration fatigue	6	-	No voltages	---	---	---	---
1042	Shock, specified pulse	7	-	Test condition K	---	---	---	---
---	Shock and vibration-fatigue test end points:							
4214	Cathode emission	9, 10	1		Δik/ik	---	10	%
4250	Power output	3, 12, 14	3	σ' = 1.3:1	po	300	---	w
4223	Mechanical tuning fatigue	-	-	No voltages; all tuners	---	200	---	cycles
---	Shelf life	20	-		t	12	---	months
---	Shelf life end point:	20d						
4250	Power output	3, 12, 14	3	σ' = 1.3:1	po	300	---	w
1105	Permanence of marking	-	-		---	---	---	---
	Quality conformance inspection, part 1	8, 13	-					
1201	Permanent shorts	---	---	Ef = 6.3 V; eb = 0	---	---	---	---
1256	Cathode current	-	1		ik	---	670	ma
4214	Cathode emission	9, 10	1		Δik/ik	---	10	%
4289	Heater current	-	-	Ef = 6.3 V; eb = 0	If	2.5	3.0	A
4253	Gain	11	2		Gss	30	---	dB
4251	Output power and gain variation	3, 12	3	Vary load through all phases:				
				Load VSWR				
				σ' = 1.3:1	po	300	---	w
				σ' = 1.5:1	po	175	---	w
				Infinite load mismatch	---	No damage		---
4223	Mechanical tuning range	5	-	High frequency	F	5,385	---	MHz
				Low frequency	F	---	5,900	MHz
4223	Mechanical tuning rate	5, 15		F1 through F3	---	---	---	---
	Input and output resonators	-	2		---	8	12	Turns
	Input and output resonators	-	-	No voltage	---	---	16	Turns between stops
	Middle resonator	-	2		---	12	16	Turns
	Middle resonator	-	-	No voltage	---	---	20	Turns between stops
4223	Operating torque	-	-	No voltage; F1 thru F3	---	---	5	in.-lbs
4223	Tuner stop endurance	-	-	No damage	---	6	---	in.-lbs

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2							
4257	Amplifier insertion loss	17	3		σ	---	2:1	---
4271	Amplifier power bandwidth	3, 12	3		ΔF	7.5	---	MHz
4266	Direct-interelec-trode capacitance	-	-	Electron gun to tube body	Cak	---	13	pF
	Quality conformance inspection, part 3							
---	Life test	13, 18, 19	3	Group D; F = F3	t	1,000	---	cycles
---	Life-test end points:							
4250	Power output	3, 12, 14	3	σ' = 1.3:1	po	240	---	w

NOTES:

1. The heater surge current shall be limited to 8.0 amperes when initially applying heater voltage. The TUT shall stabilize within the specified limits after a 2-minute (maximum) preheating period.

2. The following procedure shall be applied for new tubes and tubes which have not been operated during the previous 3-month period.

   (a) The heater shall be preheated as specified in note 1.
   (b) After the preheat period apply eb = 2.5 kv for 30 minutes, then increase eb to 5.0 kv for 15 minutes, then apply operational voltages.

3. Pulse characteristics.

   (a) Beam pulse width shall be measured between the 50 percent points of the beam current pulse when viewed by a linear circuit which presents a voltage representation of the pulse. For illustration of the pulse see MIL-STD-1311, method 4304, figure 4304-1.
   (b) The beam current pulse characteristics (when viewed as specified in note 3(a) shall be measured at the pulse amplitude points indicated below:

Pulse top:	90 percent
Rise time:	10 to 90 percent
Decay time:	90 to 10 percent

   (c) RF pulse width shall be measured between the 3 dB points of the rf output pulse when viewed through a crystal detector operating in the linear region or an equivalent circuit.
   (d) The maximum operational prf shall be determined by the maximum duty cycle and the pulse width applied.

4. Sufficient cooling shall be distributed evenly over the entire tube so that maximum temperature ratings are not exceeded. Recommended cooling at sea level, when operated at maximum power input (Pi), is 100 cfm. The maximum temperature specified for each zone shown on figure 1, shall apply.

5. A plasticized tuning curve shall be supplied with each tube. The frequency of each resonator shall be set by adjusting the appropriate tuning screws according to the tuning curve. The frequency of each resonator shall be within ±25 MHz of the frequency specified in the tuning chart. Clockwise rotation of the tuning screws shall increase the frequency of each resonator.

6. The TUT shall withstand the application of type I vibration in accordance with MIL-STD-167 except that nonresilient mountings shall be used and the amplitudes from 22 to 33 Hz shall be three times those specified. The TUT shall be mounted in the fixture shown on figure 2.

7. The TUT shall withstand the application of two blows in each of three mutually perpendicular planes or equivalent when mounted in a manner that simulates the mounting shown on figure 2 such that the TUT and mounting fixture have a natural frequency of 31 ± 3 Hz with a maximum deflection of 3/8 inch (9.53 mm).

8. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

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NOTES: -Continued

9. The heater voltage shall be decreased from 6.3 to 5.7 V. The percentage change in cathode current after 1 minute, determined inaccordance with the following formula, shall not exceed the specified limit.

$$\frac{\Delta ik}{ik}\ \% = \frac{ik\ (6.3\ v) - ik\ (5.7\ v)}{ik\ (6.3\ v)} \times 100$$

10. This test shall be performed at the conclusion of the holding period.

11. The three resonators of the TUT shall be identically tuned to the rf drive frequency at the specified eb. The peak rf output power obtained under these conditions shall be used to determine the small signal gain (Gss).

12. Tune the input and output resonator to the drive frequency specified herein and detune the middle resonator toward frequencies greater than the specified rf drive frequency until the power output is optimized for the specified rf drive power.

13. Sufficient cooling shall be distributed evenly over the entire tube so that maximum temperature ratings are not exceeded. Recommended cooling at sea level, when operated at maximum power input (Pi), is 100 cfm. The maximum temperature found in each zone shown on figure 1, shall apply.

14. The TUT shall be operable at an ambient temperature of -21°C.

15. The specified frequency range shall be covered by the number of turns specified for each resonator.

16. With the TUT in the environment chamber, adjust to obtain the test condition. Turn the TUT off and reduce the chamber temperature to the specified low value and allow to stablize for two hours. Start the TUT and measure the power output. Turn the TUT off and increase the temperature to the specified high value. Start the TUT and measure the power output. The difference in power level shall not be greater than the specified maximum limit.

17. Tune the TUT to F1, then apply test conditions. Measure the VSWR at the input connector and turn off all voltages except heater voltage. Repeat this procedure at 5,500, 5,600, 5,700, 5,800, and 5,900 MHz. The VSWR at each frequency shall not exceed the specified limit.

18. A cycle shall consist of the following events:

    (a) Operation under test conditions for 30 minutes
    (b) Standby with heater power only for 15 minutes

19. Tubes selected for this test shall have passed quality conformance inspection, part 1.

20. The TUT shall be capable of storage for 12 months in shipping containers at air temperatures within shipping containers from -54° to +71°C without impairment of subsequent performance. The TUT, furthermore, shall be capable of meeting all requirements specified herein after 50 minutes of initial warm up. The above shall be demonstrated by means of the following tests:

    (a) Cold cycle. With the TUT in the test chamber and beginning at laboratory ambient temperatures, the temperature of the chamber shall be cycled from +21° to -54°C and back to +21°C. This cycle shall be repeated five times. The -54°C temperature shall be maintained for at least 3 hours during each cycle. Also, the +21°C temperature shall be maintained for at least 3 hours between cycles.
    (b) Hot cycle. Starting at laboratory ambient temperature, the temperature of the chamber shall be cycled from +21° to +71°C and back to +21°C. This cycle shall be repeated five times. The +71°C temperature shall be maintained for at least 3 hours between cycles.
    (c) Either the hot or cold cycle may be performed first. The rate of temperature change of the test chamber from +21° to either -54° or +71°C shall be accomplished in no more than 3 hours. Likewise, the change from +71° or -54° to +21°C shall require no more than 3 hours.
    (d) After completion of the specified cycles for each of the tests, the TUT shall be stablized at ambient laboratory temperature, and tested for conformance to the specified end point listed herein under qualification testing.

21. Revision letter symbols are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

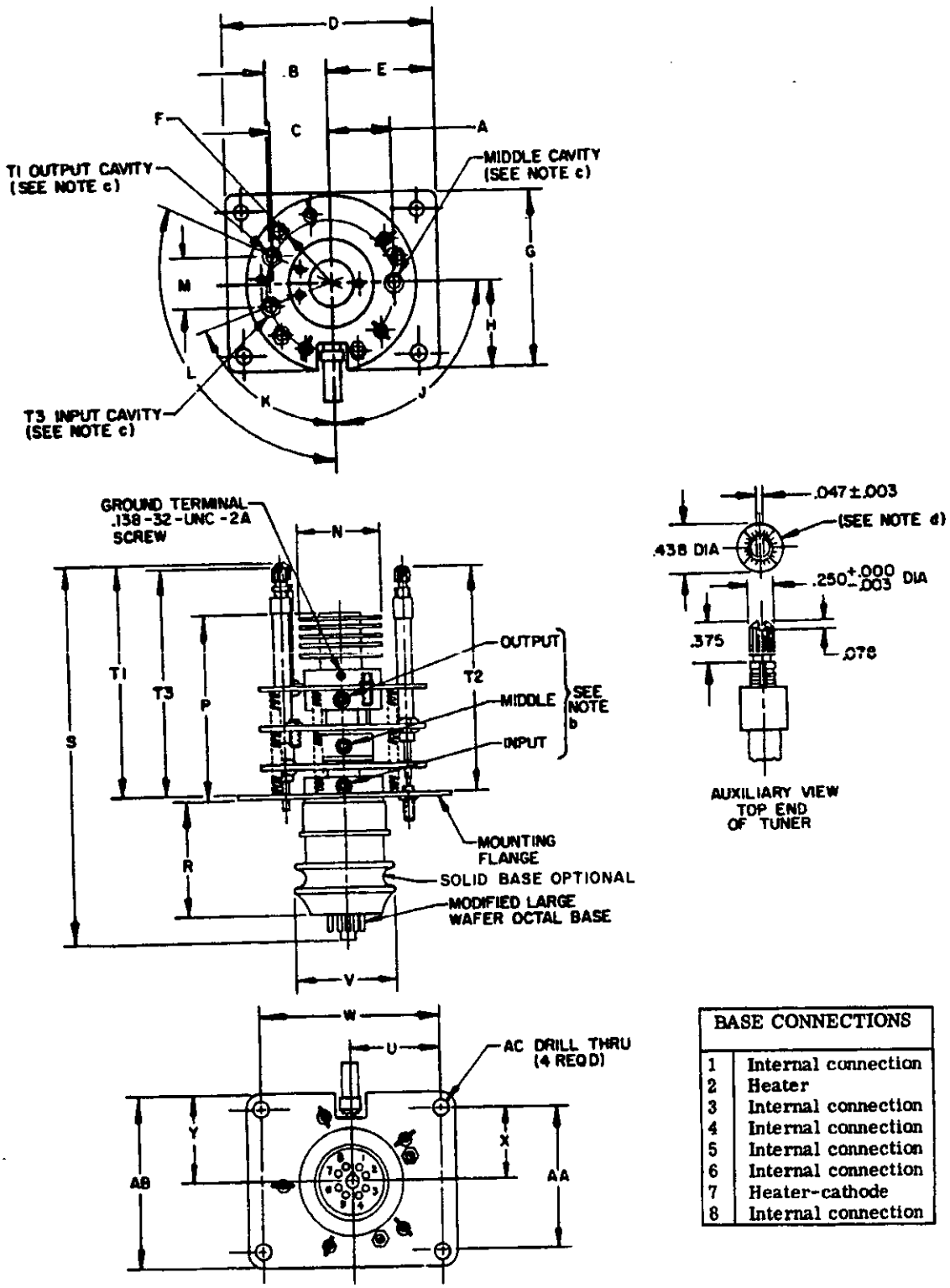
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Preparing activity:
Navy - EC

Agent:
DSA - ES

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BASE CONNECTIONS	
1	Internal connection
2	Heater
3	Internal connection
4	Internal connection
5	Internal connection
6	Internal connection
7	Heater-cathode
8	Internal connection

FIGURE 1. Outline drawing of electron tube type 8315.

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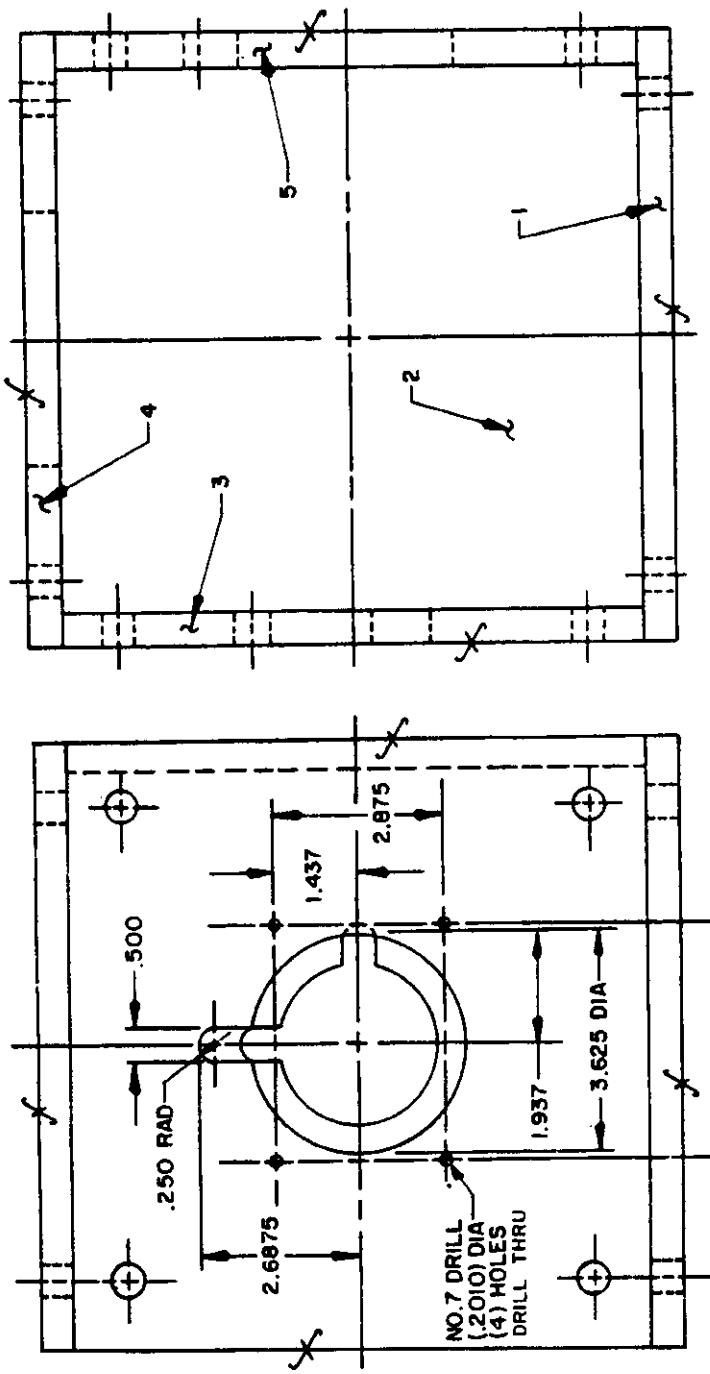
Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Note
	Minimum	Maximum	
	Quality conformance inspection, part 2		
A	1.063 (26.99)	1.438 (36.51)	a
B	1.031 (26.19)	1.563 (39.69)	a
C	.906 (23.02)	1.406 (35.72)	a
M	.875 (22.23)	---	
N	---	1.750 (44.45)	
P	---	4.375 (111.13)	
R	2.375 (60.33)	2.625 (66.68)	
V	---	2.250 (57.15)	
T1	5.063 (128.59)	5.625 (142.88)	
T2	4.969 (126.21)	5.688 (144.46)	
T3	5.000 (127.00)	5.594 (142.08)	
	Reference dimensions		
D	4.500 (114.30)		
E	2.250 (57.15)		
F	2.625 (66.68) Dia		
G	3.500 (88.90)		
H	1.750 (44.45)		
J	90°	---	
K	70°	---	
L	115°	---	
S	9.125 (231.78)		
U	1.938 (49.21)		
W	3.875 (98.43)		
X	1.438 (36.51)		
Y	2.406 (61.12)		
AA	2.875 (73.03)		
AB	4.156 (105.57)		
AC	.2187 (5.55)		

INCHES	MM
.003	.08
.047	1.19
.078	1.98
.138	3.51
.250	6.35
.375	9.53
.438	11.13

NOTES:
a. Dimensions shall apply at top of tuning shafts and collector only.
b. Mates with M39012/16-0001 (formerly UG-88E/U).
c. Spline tuning shaft shall conform to American standard B92.1-1970, class A fit, 11 teeth, 48/96 diametral pitch, 30° pressure angle.
d. Clockwise rotation of tuners shall increase the frequency.

FIGURE 1. Outline drawing of electron tube type 8315 - Continued.

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Detail No.	Material	Size	No. reqd.
5	SS BLK RED	.500 x 9.500 x 10.250	1
4	SS BLK RED	.500 x 10.250 x 10.250	1
3	SS BLK RED	.500 x 9.500 x 10.250	1
2	SS BLK RED	.500 x 9.500 x 9.500	1
1	SS BLK RED	.500 x 10.250 x 10.250	1

INCHES	MM
.2010	5.11
.250	6.35
.500	12.70
1.437	36.50
2.6875	68.26
2.875	73.03
3.625	92.08
3.875	98.43
9.500	241.30
10.250	260.35

NOTE: All welded construction.

FIGURE 2. Test fixture for electron tube type 8315.

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